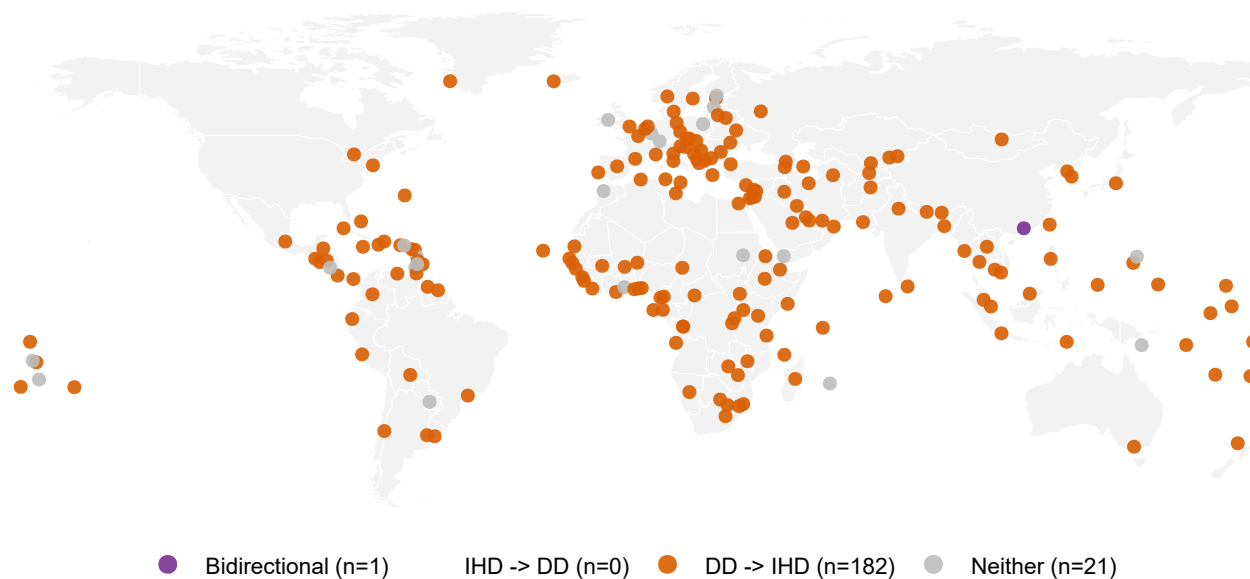


Corrected SDI-adjusted directional Granger-type analysis, 1992-2021

Annual log changes; separate target equations ensure each direction tests only the intended outcome

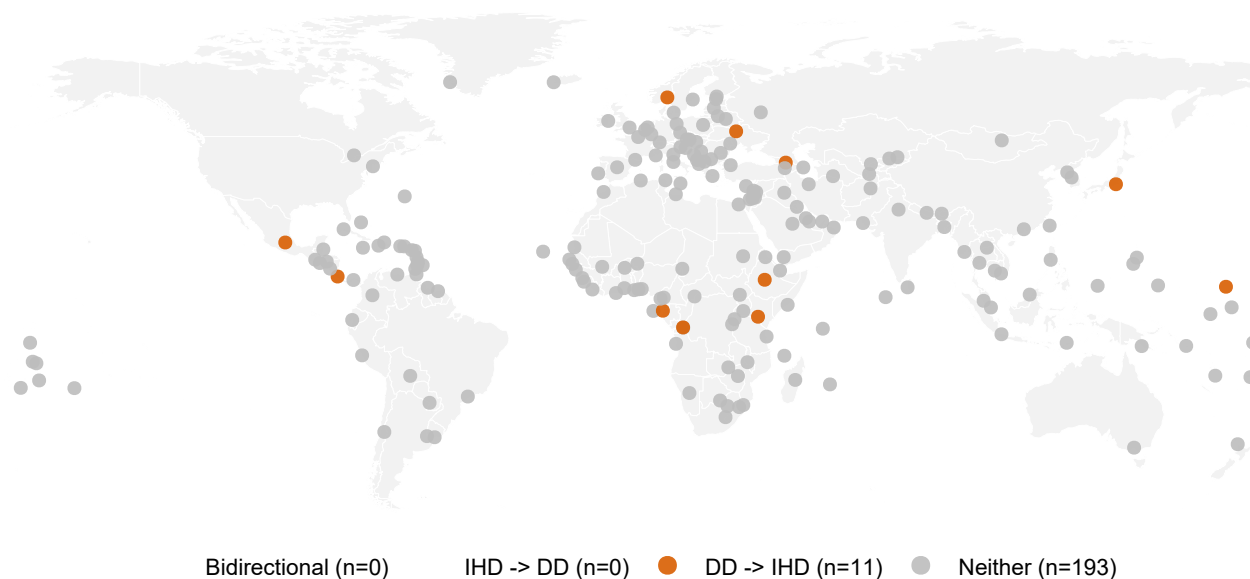
A Primary Newey-West classification (exploratory)

BIC-selected lags 1-3; SDI exogenous; Newey-West(2); directional BH $q < 0.05$



B Covariance-estimator sensitivity

Same models and BH families; HC3 covariance



Country-level temporal prediction, not individual-level or mechanistic causation. Representative coordinate points show all 204 units, including small islands. At least one residual serial-correlation flag occurred in 84/204 country analyses (90/408 directional equations); all 408 directional equations had an observation above Cook's $D > 4/n$.